# MILITARY SPECIFICATION

## READ SWITCH: 10526456

## 1. SCOPE

1.1 This specification covers one type of read switch, part of the Gun Direction Computer: M18. The read switch is used to select channels for reading information from the memory of the computer.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel, General Specification Governing the Manufacture and Inspection of |
| MIL-P-14232 | Parts, Equipment and Tools for Ordnance Materiel: Packaging of |
| MIL-I-45607 | Inspection Equipment, Supply and Maintenance of |
| MIL-C-46357 | Computer, Gun Direction: M18 Manufacturing Processes and Inspection of |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | Quality Assurance Terms and Definitions |
| MIL-STD-202 | Test Methods for Electronic and Electrical Component Parts |

DRAWINGS

U.S. Army Munitions Command

| | |
|---|---|
| F10526456 | Assembly, Read Switch |

PACKAGING DATA SHEET

| | |
|---|---|
| M14002 | Data Sheet, Master Packaging |

FSC 1220

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(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Fabrication.- The read switch shall be manufactured in accordance with Drawing F10526456 and drawings pertaining thereto.

3.2 General specifications.- The read switch shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerance
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357.

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The read switch shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the read switch for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the read switch shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The read switch shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The read switch shall meet the requirements of 3.4.1.6 and 3.4.5 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The read switch shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.



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3.3.4 Humidity.

3.3.4.1 Operating.- The read switch shall meet the requirements of 3.4.1.6 and 3.4.5 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of +135°F.

3.3.4.2 Non-operating.- The read switch shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The read switch shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The read switch shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine mist of a 5%salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the read switch subassembly shall meet the performance requirements of this specification at standard ambient temperatures between 60° and 90°F and at a barometric pressure of 28 to 32 inches of mercury with a RH up to 90%.

3.4.1 Switch circuits.- The following requirements shall apply for all six (6) identical switch circuits included on the Read Switch circuit board (see 6.1).

3.4.1.1 Transistor (collector leakage current).- The read switch transistor collector leakage current shall be less than 15 microamps (ua), with a 3 volt collector to base voltage.

3.4.1.2 Transistor (base-emitter voltage).- The read switch transistor base-emitter voltage shall be 0.3 ± 0.1 volt.

3.4.1.3 Series diode (forward voltage drop). The read switch series diodes shall have a forward voltage drop of 0.6 ± 0.1 volt, when forward biased.

3.4.1.4 Control diode (forward voltage drop).- The read switch control diodes shall have a forward voltage drop of less than 0.6 volt at a current level of 1.8 ma.

3.4.1.5 Control diode (leakage current).- The read switch control diodes shall have a leakage current of less than 30 ua with 6 volts reverse voltage.

3.4.1.6 Read switch circuit (voltage gain).- The read switch shall have a signal voltage gain of greater than 0.9 volt.

3.4.2 Amplifier circuit.

3.4.2.1 Quiescent operation.- All transistors in the amplifier circuit shall be biased for linear (Class A) operation.



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3.4.2.2 A.C. voltage gain.- The amplifier circuit shall have an a.c. voltage gain of 29.5 plus 4 or minus 6 percent.

3.4.2.3 Threshold setting.- The amplifier circuit shall have a threshold value of 45.0 ± 1.3 millivolts (mv).

3.4.3 Flip-flop (steady state).- The saturation voltages of the "on" transistors must not be greater than 0.3 volt and the "off" transistors must have at least 0.3 volt, but not greater than 1.5 volts, reverse bias from base to emitter.

3.4.4 Miscellaneous diodes.

3.4.4.1 Germanium diodes.- The germanium diodes shall have less than 0.6 volt drop when conducting and a leakage current of less than 10 ua.

3.4.4.2 Silicon diodes.- The silicon diodes shall have less than 0.7 volt drop when conducting and a leakage current of less than 1 ua.

3.4.5 Flip-flop and strobe switch (dynamic operation).- The flip-flop and strobe switch shall be capable of switching states at a frequency of 250 kc ± 10 percent.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.



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4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirement and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I

(100 percent inspection)

| Characteristic | Requirement | Test Procedures |
|---|---|---|
| 101. Transistor (Collector leakage current) | 3.4.1.1 | 4.5.2.1.1 |
| 102. Transistor (base-emitter voltage) | 3.4.1.2 | 4.5.2.1.2 |
| 103. Series diode (forward voltage drop) | 3.4.1.3 | 4.5.2.1.3 |
| 104. Control diode (forward voltage drop) | 3.4.1.4 | 4.5.2.1.4 |
| 105. Control diode (leakage current) | 3.4.1.5 | 4.5.2.1.5 |
| 106. Read switch circuit (voltage gain) | 3.4.1.6 | 4.5.2.1.6 |
| 107. Quiescent operation | 3.4.2.1 | 4.5.2.2.1 |
| 108. A.C. voltage gain | 3.4.2.2 | 4.5.2.2.2 |
| 109. Threshold setting | 3.4.2.3 | 4.5.2.2.3 |
| 110. Flip-flop (steady state) | 3.4.3 | 4.5.2.3, 4.5.2.3.1 |
| 111. Flip-flop and strobe switch (dynamic operation) | 3.4.5 | 4.5.2.4 |
| 112. Miscellaneous diodes (germanium and silicon) | 3.4.4.1, 3.4.4.2 | 4.5.2.5 |

4.3.1 General.- One read switch shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

| CONTROL TEST | REQUIREMENT | TEST PROCEDURES |
|---|---|---|
| 301. Fabrication | 3.1 | Applicable drawings - visual |
| 302. General specification | 3.2 | MIL-F-13926 and MIL-C-46357 - visual |

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.



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TABLE III

| CONTROL | REQUIREMENT | TEST PROCEDURES |
|---|---|---|
| 303. Vibration | 3.3.1 | 4.5.1.1 |
| 304. Shock | 3.3.2 | 4.5.1.2 |
| 305. Humidity-Operating | 3.3.4.1, 3.4.1.6, 3.4.5 | 4.5.1.3.2, 4.5.2.1.6.2 |
| 306. Humidity-Non-operating | 3.3.4.2 | 4.5.1.3.1 |
| 307. Operating temperature | 3.3.3.1, 3.4.1.6, 3.4.5 | 4.5.1.5, 4.5.2.1.6.2 |
| 308. Storage temperature | 3.3.3.2 | 4.5.1.5 |
| 309. Salt spray | 3.3.6 | 4.5.1.4 |

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

| Item Number | Description |
|---|---|
| 1 | Voltage measuring device-alternating current: frequency range 10 cycle per second (cps) to 4 megacycles (mc), voltage range 0.1 millivolts (mv) to 300 volts, accuracy ± 1 percent. Input impedance 10 megohms (Mohms) and 15 pico farads. |
| 2 | Voltage measuring device-direct current: accuracy ± 5 percent, voltage range 10mv to greater than 100 volts. Input resistance greater than or equal to 100K ohm. |



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| Item Number | Description |
|---|---|
| 3 | Oscilloscope: accuracy ± 4 percent, rise time 50 nanoseconds or less, sweep shall be from 0.1 microseconds (us) per centimeter to 1 ms per centimeter or greater, input resistance greater than or equal to 1 megohm, input capacitance 20 pico farads less. |
| 4 | Attenuating probe: attenuating X10. Input resistance 10 Mohms or greater, input capacitance 10 pico farads or less. |
| 5 | Sine wave generator: frequency 250 kc ± 10 percent, output voltage 0 to 500mv peak-to-peak or greater, output resistance less than 700 ohm. Distortion less than 3 percent. |
| 6 | Power Supply: minus 18 volts ± 2 percent, current rating greater than or equal to 70ma, peak-to-peak ripple equal to or less than 600mv, output impedance equal to or less than 2 ohm from dc to 500 kc. |
| 7 | Power Supply: minus 10 volts ± 2 percent, current rating equal to or greater than 25ma, peak-to-peak ripple equal to or less than 100mv, output impedance equal to or less than 2 ohm from dc to 500 kc. |
| 8 | Power Supply: minus 6 volts ± 2 percent, current rating equal to or greater than 25ma, peak-to-peak ripple equal to or less than 400mv, output impedance equal to or less than 2 ohms from dc to 500 kc. |
| 9 | Power Supply: minus 3 volts ± 2 percent, current rating equal to or greater than ± 10ma, peak-to-peak ripple equal to or less than 100mv, output impedance equal to or less than 2 ohms from dc to 500 kc. |
| 10 | Power Supply: plus 1.25 volts ± 2 percent, current rating equal to or greater than 15ma, peak-to-peak ripple equal to or less than 100mv, output impedance equal to or less than 2 ohms from dc to 500 kc. |
| 11 | Power Supply: plus 6 volts ± 2 percent, current rating equal to or greater than 10ma, peak-to-peak ripple equal to or less than 100 mv, output impedance equal to or less than 2 ohms from dc to 500 kc. |

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of



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measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the read switch rigidly mounted in its normal operating position. Subsequent to the vibration test, the read switch shall show no damage or defect resulting from the test when examined by visual and tactile means. The read switch shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the read switch shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that receives the read switch in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F 13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.1.6.2 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The read switch shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the read switch shall be inspected for any physical damage, then tested in accordance with 4.5.2.6.2 and 4.5.4. The read switch shall subsequently be inspected as specified in 4.5.1.5 prior to determining compliance with 3.4 as required by 3.3.6.

4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the read switch shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the read switch shall be tested in accordance with 4.5.2.1.6.2 to determine compliance with the requirements of 3.4.3 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the read switch shall be tested in accordance with 4.5.2 to 4.5.5 inclusive to determine compliance with 3.3.3.2.



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4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 read switches produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.

4.5.2.1 Switch circuit tests.

4.5.2.1.1 Transistor (collector leakage current).- To determine compliance with 3.4.1.1, connect the minus 3 volt power supply to pin 43A and the ground return of this power supply to pin 40A. The voltage measured from each of the pins specified in Column A of Table IV when measured to the respective pin of Column B of Table IV shall be less than plus 0.27 volts.

TABLE IV

| Column A | Column B |
|---|---|
| 46A | 48A |
| 49B | 51B |
| 49A | 51A |
| 46B | 48B |
| 56B | 58B |
| 56A | 58A |

4.5.2.1.2 Transistor (base-emitter voltage).- To determine compliance with 3.4.1.2 connect the minus 18 volt power supply to pin 24A and the ground return of this power supply to pin 39A. The voltage measured from each of the pins or test points specified in Column A of Table V when measured to the respective pin or test points of Column B of Table V shall be plus 0.2 ± 0.1 volts.

TABLE V

| Column A | Column B |
|---|---|
| 48A | Tp1 |
| 51B | Tp2 |
| 51A | Tp3 |
| 48B | Tp12 |
| 58B | Tp13 |
| 58A | Tp14 |

4.5.2.1.3 Series diode (forward voltage drop).- The forward voltage drop requirement in 3.4.1.3 shall be determined by the tests in 4.5.2.1.3.1 to 4.5.2.1.3.3 inclusive.

4.5.2.1.3.1 The voltage measured from each of the pins or test points specified in Column A of Table VI when measured to the respective pins or test points of Column B of Table VI shall be 0.6 ± 0.10 volts.



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TABLE VI

| Column A | Column B | Column A | Column B |
|---|---|---|---|
| Tp1 | Tp4 | Tp12 | Tp15 |
| Tp1 | Tp5 | Tp12 | Tp16 |
| Tp2 | Tp6 | Tp13 | Tp17 |
| Tp2 | Tp7 | Tp13 | Tp18 |
| Tp3 | Tp8 | Tp14 | Tp19 |
| Tp3 | Tp9 | Tp14 | Tp20 |

4.5.2.1.3.2 Connect a 4.7K ohm ± 1 percent, 1/4 watt carbon film resistor from pin 65B to pin 40A. The following six tests are to be made in sequence. Connect the pins specified in Column A of Table VII to pin 39A and measure the voltage from Tp10 to the respective test point of Column B of Table VII. The voltage measured shall be 0.6 ± 0.10 volts. At the completion of each test the connections to pin 39A will be removed.

TABLE VII

| Test Number | Column A | Column B |
|---|---|---|
| 1 | 55A, 59A, 61B, 62B, 44A | Tp5 |
| 2 | 45B, 59A, 44A, 61B, 62B | Tp7 |
| 3 | 45B, 55A, 44A, 61B, 62B | Tp9 |
| 4 | 45B, 55A, 59A, 61B, 62B | Tp16 |
| 5 | 45B, 55A, 59A, 44A, 62B | Tp18 |
| 6 | 45B, 55A, 59A, 44A, 61B | Tp20 |

4.5.2.1.3.3 Connect a 4.7K ohm ± 1 percent, 1/4 watt carbon film resistor from pin 65A to pin 40A. The following six tests are to be made in sequence. Connect the pins specified in Column A of Table VIII to pin 39A and measure the voltage from Tp11 to the respective test point of Column B of Table VIII. The voltages measured shall be 0.6 ± 0.10 volts. At the completion of each test the connections to pin 39A will be removed.

TABLE VIII

| Test Number | Column A | Column B |
|---|---|---|
| 1 | 52A, 59B, 45A, 61A, 62A | Tp4 |
| 2 | 44B, 59B, 45A, 61A, 62A | Tp6 |
| 3 | 44B, 52A, 45A, 61A, 62A | Tp8 |
| 4 | 44B, 52A, 59B, 61A, 62A | Tp15 |
| 5 | 44B, 52A, 59B, 45A, 62A | Tp17 |
| 6 | 44B, 52A, 59B, 45A, 61A | Tp19 |

4.5.2.1.4 Control diode (forward voltage drop).- The forward voltage drop requirement in 3.4.1.4 shall be determined by the tests in 4.5.2.1.4.1 and 4.5.2.1.4.2.

4.5.2.1.4.1 Connect pins 37A, 52B, 55B and 38A to pin 40A. The voltage at each of the following test points when measured with respect to pin 40A shall be no more negative than minus 0.6 volt: Tp4, Tp5, Tp6, Tp7, Tp8, Tp9, Tp15, Tp16, Tp17, Tp18, Tp19, Tp20. Remove the connection between pins 37A, 52B, 55B, 38B and pin 40A.



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4.5.2.1.4.2 Connect pins 44B, 45B, 52A, 55A, 59B, 59A, 45A, 44A, 61A, 61B, 62A, 62B to pin 40A. The coltage at each of the following test points when measured with respect to pin 40A shall be no more negative than minus 0.6 volt: Tp4, Tp5, Tp6, Tp7, Tp8, Tp9, Tp15, Tp16, Tp17, Tp18, Tp19, Tp20. Disconnect pins 44B, 45B, 52A, 55A, 59B, 59A, 45A, 44A, 61A, 61B, 62A, 62B from pin 40A.

4.5.2.1.5 Control diode (leakage current).- Leakage current tests shall be performed in accordance with 4.5.2.1.5.1 to 4.5.2.1.5.3 inclusive to determine compliance with 3.4.1.5.

4.5.2.1.5.1 Connect the minus 6 volt power supply through a 10K ohm, ± 1%, 14/ watt resistor to the pins listed in Column A of Table IX. Connect the pins listed in Column B of Table IX to pin 40A. Measure the voltage of the pins listed in Column A with respect to the minus 6 volt power supply. The voltage shall be less than 0.5 volts. Remove the minus 6 volt and the 10K ohm resistor from the pins listed in Column A. Remove the connections between the pins listed in Column B and pin 40A.

TABLE IX

| Test Number | Column A | Column B |
|---|---|---|
| 1 | 37A | 45A |
| 2 | 37A | 44B |
| 3 | 52B | 52A |
| 4 | 52B | 59B |
| 5 | 52B | 62A |
| 6 | 52B | 61A |
| 7 | 55B | 55A |
| 8 | 55B | 59A |
| 9 | 55B | 61B |
| 10 | 55B | 62B |
| 11 | 38A | 45B |
| 12 | 38A | 44A |

4.5.2.1.5.2 Connect the minus 6 volt power supply through a 10K ohm resistor to pin 44B. Connect pins 37A, 52B, 55B, 38A to pin 40A. Measure the voltage at pin 44B with respect to the minus 6 volt power supply. The voltage shall be less than plus 0.3 volt. Remove the minus 6 volt power supply and the 10K ohm resistor from pin 44B.

5.2.1.5.3 Repeat 4.5.2.1.5.2 with pins 45B, 52A, 55A, 59B, 59A, 45A, 44A, 62A and 62B substituted for pin 44B. Then remove pins 37A, 52B, 55B and from pin 40A.

4.5.2.1.6 Read switch circuit (voltage gain) (60° to 90°F).- Voltage gain will be measured in the following tests to determine compliance with 3.4.1.3.

4.5.2.1.6.1 Connect one putput terminal of the sine wave generator (item 5 of 4.10.2) to pin 40A. Connect the other output terminal of the sine wave generator through a 0.1 microfarad ± 10 percent, 100 volt, non-polarized capacitor to the pin specified in Column A of Table X and connect the respective pins of Column B to pin 40A. Apply a sine wave, 0.2 ± 0.05 volts peak-to-peak, 250 kc ± 10 percent to the pin specified in Column A and measure the output sine waves from Tp10 to pin 40A and from Tp11 to pin 40A. The amplitude of each of these output sine waves shall be greater than 0.9 times the amplitude of the input sine wave.



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Apply a signal to the pin in Column A and connect the pins of Column B to pin 40A.

TABLE X

| Test Number | Column A | Column B |
|---|---|---|
| 1 | 48A | 52A, 55A, 59B, 59A, 45A, 44A, 61A, 61B, 62A, 62B |
| 2 | 51B | 44B, 45B, 59B, 59A, 45A, 44A, 61A, 61B, 62A, 62B |
| 3 | 51A | 44B, 45B, 52A, 55A, 45A, 44A, 61A, 61B, 62A, 62B |
| 4 | 48B | 44B, 45B, 52A, 55A, 59B, 59A, 61A, 61B, 62A, 62B |
| 5 | 58B | 44B, 45B, 52A, 55A, 59B, 59A, 45A, 44A, 62A, 62B |
| 6 | 58A | 44B, 45B, 52A, 55A, 59B, 59A, 45A, 44A, 61A, 61B |

4.5.2.1.6.2 Read switch circuit (voltage gain ), flip-flop and strobe switch dynamic operation (-40° and +135°).- To determine compliance with 3.4.1.6 and 3.4.5 as required by 3.3.3.1 and 3.3.4.1 the following tests shall be conducted.

4.5.2.1.6.2.1 Connect the minus 18 volt power supply to pin 25A. Connect the minus 3 volt power supply to pin 43A. Connect the power grounds of these supplies to pin 39A and connect pin 40A to pin 36A. Connect a 4.7K ohm ± 1 percent, 1/4 watt, carbon film resistor from pin 65B to pin 40A. Connect a 4.7K ohm ± 1 percent, 1/4 watt, carbon film resistor from pin 65A to pin 49A. Repeat paragraph 4.5.2.1.6.1. Remove all external connections to the subassembly.

4.5.2.1.6.2.2 Connect the plus 6 volt power supply to pin 25B. Connect the plus 1.25 volt power supply to pin 7A. Connect the minus 6 volt power supply to pins 29A and 30A. Connect the minus 10 volt power supply to pins 1A and 13B. Connect the power grounds of the above power supplies to pins 39A, 40A, 36A, 35A, 7B and 8B. Repeat 4.5.2.4. Remove all external connections to the subassembly.

4.5.2.2 Amplifier circuit tests.

4.5.2.2.1 Quiescent operation.- To determine compliance with 3.4.2.1, connect the plus 6 volt power supply to pin 25B and the ground return to pin 35A. Connect the minus 3 volt power supply to pin 20A and 43A and the ground returns to pin 35A and pin 39A. Connect the minus 6 volts power supply to pin 29A and the ground return to pin 36A. Measure the dc voltage from the pin or test point specified in Column A of Table XI to pin 36A. The voltage measured shall be the same as that specified in the same row of Column B of Table XI.

TABLE XI

| Column A | Column B |
|---|---|
| Pin 41B | -2.8 ± 0.5 volts |
| Tp25 | -2.9 ± 1.3 volts |
| Tp26 | -5.0 ± 2.0 volts |
| Tp27 | -2.7 ± 1.3 volts |
| Tp28 | -7.7 ± 2.0 volts |
| Tp29 | -4.8 ± 2.0 volts |
| Tp31 | -7.5 ± 2.0 volts |



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TABLE XI (Continued)

| Column A | Column B |
|---|---|
| Tp34 | -11.0 ± 2.0 volts |
| Tp37 | + 0.5 ± 0.5 volts |
| Tp40 | - 3.0 ± 1.5 volts |
| Pin 5B | -12.5 ± 2.0 volts |
| Tp42 | -12.3 ± 2.0 volts |

4.5.2.2.2 A.C. voltage gain.- To determine compliance with 3.4.2.2, connect one output terminal of a sine wave generator to pin 36A. Connect the other terminal of the sine wave generator through a 0.1 microfarad ± 10 percent, 100 volt, non-polarized capacitor to pin 41B. Apply a 0.2 ± 0.05 volt, 250 kc ± 10 percent sine wave to pin 41B. Measure the output sine wave from Tp35 to pin 36A. The output sine wave amplitude shall be at least 27.7 and not greater than 30.7 times the input sine wave amplitude.

4.5.2.2.3 Threshold setting.- To determine compliance with 3.4.2.3 use the power supply connections specified in 4.5.2.2.1. Connect the sine wave generator (Item 5 of 4.10.2) directly to pin 39B and through a 0.1 microfarad, ± 10 percent, 100 volt non-polarized capacitor to pin 41B. Connect the oscilloscope (Item 5 of 4.10.2) using the probe (Item 4 of 4.10.2) to pin 12B. Connect the oscilloscope ground to pin 35A. Connect the voltage measuring device (Item 1 of 4.10.2) to pins 39B and 41B. Observe the waveform on the oscilloscope. Select resistor R64 from Table XII so that when the input sine wave is increased from zero amplitude the output waveform is less than 0.90 volts, zero to peak until the input signal reaches 45.0 ± 1.3 mv zero to peak. At this input voltage one of the waveforms shown in Figure 1 will appear on the oscilloscope.

TABLE XII

| Ohms | Part Number | Ohms | Part Number |
|---|---|---|---|
| 1.24K | 10526091 | 6.2K | RN60C6201F |
| 2.2 K | 10526094 | 6.8K | 10526100 |
| 2.7 K | 10526095 | 7. K | RN60C7001F |
| 3.3 K | 10525856 | 7.2K | RN60C7201F |
| 3.9 K | 10526096 | 7.4K | RN60C7401F |
| 4.32K | 10526097 | 7.5K | RN60C7501F |
| 4.7 K | 10526098 | 7.8K | RN60C7801F |
| 5.1 K | RN60C5101F | 8. K | RN60C8001F |
| 5.3 K | RN60C5301F | 8.2K | RN60C8201F |
| 5.5 K | RN60C5501F | 9.1K | 10526598 |
| 5.6 K | 10526099 | 10. K | 10526599 |

4.5.2.3 Flip-flop (steady state).- Compliance with 3.4.3 shall be determined by connecting the plus 1.25 volt power supply to pin 7A and the minus 10 volt power supply to pin 13B. Connect the ground returns of the above power supplies to pins 7B and 8B. Connect the minus 6 volt power supply to pin 19B momentarily. Measure the voltage from the pin indicated in column A of Table XIII to the corresponding pin in column B of Table XIII. The voltage is indicated in column C of Table XIII.



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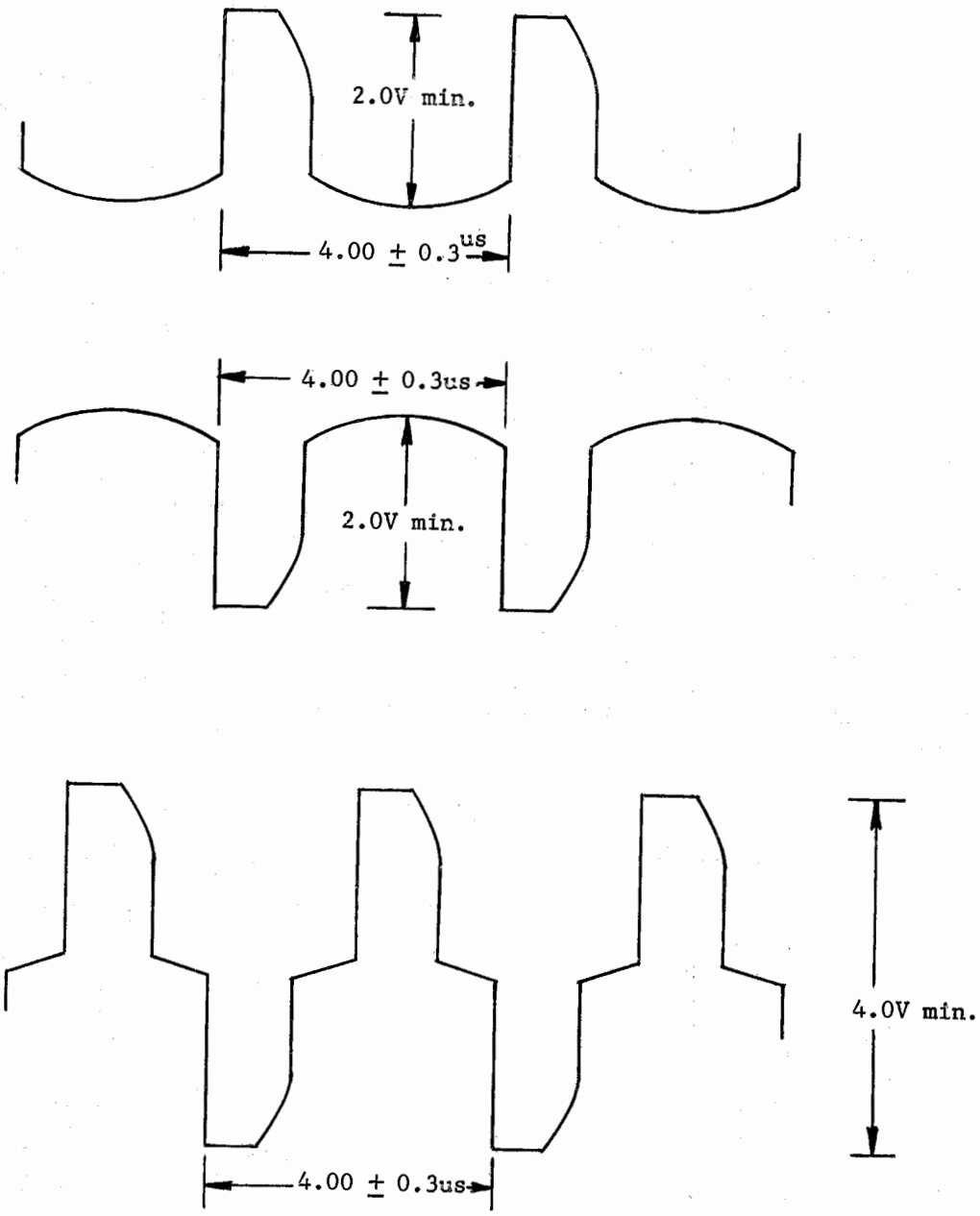



FIGURE 1



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TABLE XIII

| Column A | Column B | Column C |
|---|---|---|
| Pin 7A | Pin 4B | Less than +0.3V DC |
| 7B | 6B | Less than +0.3V DC |
| 5A | 30A 30A | Less than +20 millivolts DC |
| 4A | 7A | +0.53 ± 0.2V DC |
| 19A | 1 14A | +1.3 ± 0.2V DC |

4.5.2.3.1 Connect the minus 6 volt power supply to pin 3A momentarily. Measure the voltage from the pin indicated in Column A to Table XIV to the corresponding pin in Column B of Table XIV. The voltage is indicated in Column C of Table XIV.

TABLE XIV

| Column A | Column B | Column C |
|---|---|---|
| 7A | 8A | Less than +0.3V DC |
| 7B | 5A | Less than +0.3V DC |
| 30A | 6B | +0.65 ± 0.1V DC |
| 10A | 7A | +0.53 ± 0.2V DC |
| 14A | 19A | +1.3 ± 0.2V DC |
| 7B | 14A | +3.6V ± 0.5V DC |

4.5.2.4 Flip-flop and strove switch (dynamic operation).- Compliance with 3.4.5 shall be determined by connecting pins 8B, 35A and 40A together. Connect the minus 10 volt power supply to pin 1A with ground return to pin 8B. Connect one output terminal of the sine wave generator (Item 5 of 4.10.2) directly to pin 39B. Connect the other output terminal through a 0.1 microfarad, ± 10 percent, 100 volts, non-polarized capacitor to pin 41B. Apply a 0.2 plus or minus 0.05 volt, 250 KC plus or minus 10 percent sine wave to pin 41B. Connect the oscilloscope (Item 3 of 4.10.2) using the probe (Item 4 of 4.10.2) to pin 6B. Measure 5A the same as 6B. Connect the oscilloscope ground to pin 35A. The waveform observed on the oscilloscope shall meet the requirements of Figure 2.

4.5.2.5 Miscellaneous diodes (Germanium and Silicon).- Disconnect -10V from pin 1A. The following connections and measurements shall be made to determine compliance with 3.4.4.1 and 3.4.4.2:

(a) Connect the minus 18 volt power supply to pin 18A.

(b) Connect minus 6 volts to pin 3A momentarily. Measure the voltage from pin 5A to Tp38. The voltage shall be less than .6 volt.

(c) Connect pin 15A to pin 7B. Measure the voltage from pin 15A to Tp32. It shall be less than 0.7 volts. Remove connection between 15A and 7B.

(d) Connect Tp33 to pin 7B. Measure the voltage from 7B to Tp32. The voltage shall be less than .6 volt.

(e) Connect the plus 6 volt power supply through a 10K ohm, 5 percent, 1/4W resistor to pin 2A. Measure the voltage from pin 2A to pin 7B. The voltage shall be less than .6 volt. Remove connection to pin 2A.



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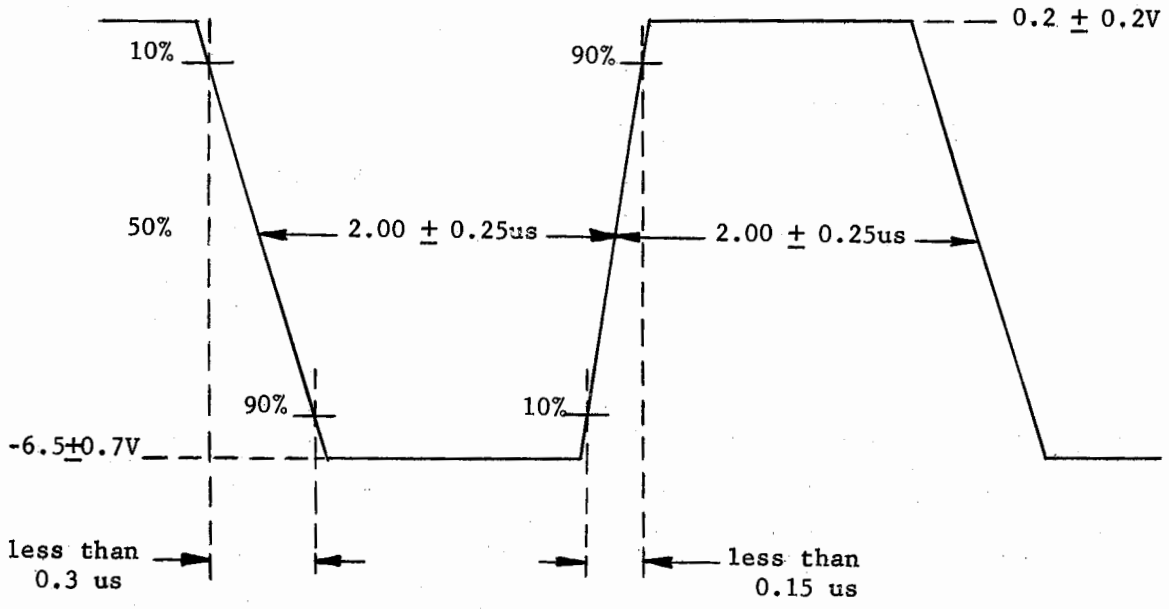



FIGURE 2



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(f) Connect the plus 6 volt power supply through a 10K ohm, ± 5 percent, 1/4W resistor to pin 17B. Measure the voltage from pin 17B to pin 7B. The voltage shall be less than .6 volt. Remove connection to pin 17B.

(g) Connect the minus 6 volt power supply through a 10K ohm, ± 5 percent, 1/8W resistor to pin 2A. Measure the voltage from pin 2A to the minus 6 volt power supply. The voltage shall be less than 0.10 volt. Remove the connection to pin 2A.

(h) Connect the minus 6 volt power supply through a 10K ohm, ± 5 percent, 1/8W resistor to pin 17B. Measure the voltage from pin 17B to the minus 6 volt power supply. The voltage shall be less than 0.10 volt. Remove the connection from the minus 6 volt power supply. Remove the connection between Tp33 and pin 7B.

(i) Connect the minus 6 volt power supply to pin 19B momentarily. Measure the voltage from pin 6B to Tp32. The voltage shall be less than 0.6 volt.

(j) Connect pin 2B to pin 7B. Measure the voltage from pin 7B to Tp38. The voltage shall be less than 0.7 volts. Remove the connection between pin 2B and 7B.

(k) Connect Tp33 to 7B. Measure the voltage from pin 7B to Tp38. The voltage shall be less than 0.6 volts. Remove all external connections to the subassembly.

(l) Connect the plus 6 volt power supply to pin 18A. Connect the power supply ground to pin 7B. Connect Tp33 to pin 7B. Measure the voltage from pin 18A to Tp32. The voltage shall be less than 0.4 volt. Measure the voltage from pin 18A to Tp38. The voltage shall be less than 0.4 volt. Remove the connection between Tp33 and pin 7B. Remove the plus 6 volt power supply from pin 18A.

(m) Connect the minus 6 volt power supply through a 68K ohm, ± 5%, 1/4 watt resistor to pin 15A. Ground TP 32. Measure the voltage from Pin 15A to the minus 6 volt supply. The voltage shall be less than 0.04 volt. Remove the minus 6 volt power supply from pin 15A. Connect the minus 6 volt power supply through the 68K ohm resistor to pin 2B. Ground TP 38. Measure the voltage from Pin 2B to the minus 6 volt supply. The voltage shall be less than 0.04 volt. Remove the minus 6 volt power supply from Pin 2B.

(n) Connect the plus 6 volt power supply to Pin 18A. Connect pins 5A and 6B to pin 7B. Measure the voltage from pin 18A to Tp32. The voltage shall be less than 0.4 volt. Measure the voltage from pin 18A to Tp38. The voltage shall be less than 0.4 volt. Remove the connection between pin 5A, 6B and 7B. Remove the plus 6 volt power supply from Pin 18A.

(o) Connect minus 6 volt power supply to pins 19B and 3A. Connect the power supply ground to pin 7B. Connect Tp33 to pin 7B. Measure the voltage from Tp33 to Tp30. It shall be less than 0.6 volt. Measure the voltage from Tp33 to Tp23. It shall be less than 0.6 volt. Remove the minus 6 volts from pins 19B and 3A. Remove the connection between Tp33 and pin 7B. Remove the plus 1.25 volt supply from Pin 7A.



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(p) Connect the minus 6 volt power supply through a 4.7K ohm, ± 5 percent, 1/4W resistor to Tp33. Connect Tp30 and Tp23 to pin 7B. Measure the voltage from Tp33 to the minus 6 volt power supply. The voltage shall be less than 0.10 volt. Remove the connection between the minus 6 volt power supply and Tp33. Remove the connections between Tp30, Tp23 and Pin 7B.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging, and packing shall be in accordance with Master Packaging Data Sheet M14002.

## 6. NOTES

6.1 Intended use.- The Read Switch circuit board has two separate functions. The six read switches are used to select the channel to be read from the main memory. The signals then go to the Read Amplifier ciruit board where they are amplified and converted to logical signals. The amplifier, strobe switch, and flip-flop circuits on the Read Switch circuit board are used to amplify and convert the signals from the circulating registers of the memory to logical signals.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet (See 5.1).
(e) A block of serial numbers to identify each Read Switch.

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# SPECIFICATION ANALYSIS SHEET

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**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

| ORGANIZATION | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
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